

UNIT – II

Partial Derivatives

1. If $u = \log(x^2 + y^2 + z^2)$ then prove that

$$(a) \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{2}{x^2 + y^2 + z^2}$$

$$(b) x \frac{\partial^2 u}{\partial y \partial z} = y \frac{\partial^2 u}{\partial x \partial z} = z \frac{\partial^2 u}{\partial x \partial y}$$

2. If $u = \log(ax + by)$ then find $\frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial y^2}, \frac{\partial^2 u}{\partial x \partial y}, \frac{\partial^2 u}{\partial y \partial x}$

3. If $u = \log(\tan x + \tan y + \tan z)$, prove that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$

4. If $r^2 = x^2 + y^2$, then prove that $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r} \left[\left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 \right]$ (Model May 2017, May 2015)

5. If $u = \sin^{-1} \left(\frac{y}{x} \right)$, verify that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ (Model Nov. 2016)

6. If $z = f(x + ct) + g(x - ct)$, prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$

7. If $z = e^{ax+by} f(ax - by)$, show that $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$

Total Differentiation

1. Find $\frac{du}{dx}$, if $u = \cos(x^2 + y^2)$, where $a^2 x^2 + b^2 y^2 = c^2$.

2. Find $\frac{du}{dx}$, if $u = \tan^{-1} \left(\frac{x}{y} \right)$, where $x^2 + y^2 = a^2$.

3. If $u = f(x - y, y - z, z - x)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

4. If $z = f(x, y)$, where $x = e^u + e^{-v}$ and $y = e^{-u} - e^{-v}$, then prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$.

5. If $z = u^2 + v^2; x = u^2 - v^2; y = uv$ find $\frac{\partial z}{\partial x}$.

6. If $u = f(x^2 + 2yz, y^2 + 2zx)$ then prove that $(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0$.
7. If $u = f(r, s, t)$ and $r = \frac{x}{y}$; $s = \frac{y}{z}$; $t = \frac{z}{x}$; then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.
8. If $f(cx - az, cy - bz) = 0$, show that $ap + bq = c$, where $p = \frac{\partial z}{\partial x}$; $q = \frac{\partial z}{\partial y}$.
9. If $z = f(x, y)$ and u, v are the other two variables such that $u = lx + my$; $v = ly - mx$ show that, $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2 + m^2) \left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \right)$.

Polar form

1. if $u = f(x, y)$, where $x = e^r \cos \theta$; $y = e^r \sin \theta$ then prove that
- (a) $x \frac{\partial u}{\partial \theta} + y \frac{\partial u}{\partial r} = e^{2r} \frac{\partial u}{\partial y}$.
- (b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-2r} \left[\frac{\partial^2 u}{\partial r^2} + \frac{\partial^2 u}{\partial \theta^2} \right]$.
- (c) $\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = e^{-2r} \left[\left(\frac{\partial u}{\partial r} \right)^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 \right]$.
2. If $u = f(x, y)$, where $x = r \cos \theta$; $y = r \sin \theta$ then prove that (model Nov. 2016)
- (a) $\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2$.
- (b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$
3. If $x = r \cos \theta$, $y = r \sin \theta$, prove that the equation $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ is equivalent to
- $$\frac{\partial u}{\partial r} + \frac{1}{r} \tan \left(\frac{\pi}{4} - \theta \right) \frac{\partial u}{\partial \theta} = 0.$$
4. If $u = f(r)$; $x = r \cos \theta$, $y = r \sin \theta$, then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$.

Implicit function

1. If $y^3 - 3ax^2 + x^3 = 0$, Prove that $\frac{d^2y}{dx^2} + \frac{2a^2x^2}{y^5} = 0$.
2. If $(\cos x)^y = (\sin y)^x$, find $\frac{dy}{dx}$.
3. If $u = \log(xy)$ and $x^3 + y^3 + 3axy = 0$, find $\frac{du}{dx}$.
4. Find $\frac{dy}{dx}$, when $x^y + y^x = \beta$, β is constant $x > 0$ & $y > 0$.
5. Find $\frac{dy}{dx}$, when $f(x, y) = \log(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right) = 0$.

Euler's theorem Problems (or) Homogeneous function

1. Verify: Euler's theorem for $Z = [x^2 + xy + y^2]^{-1}$.
2. Show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u \log u$, where $\log u = \frac{x^3 + y^3}{3x + 4y}$.
3. If $u = \sin^{-1} \left[\frac{x+y}{\sqrt{x} + \sqrt{y}} \right]$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$.
4. If $u = \tan^{-1} \left[\frac{x^3 + y^3}{x - y} \right]$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.
5. If $u = \sin^{-1} \left[\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right]$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

Type I

Taylor's Series

❖ Given point value (x, y)

Find the Taylor's Series Expansion for the following

1. $f(x) = x^2y + 3y - z$ about the point $(1, -2)$ upto 2nd degree. (Model Nov. 2016, Jan. 2016).
2. $f(x) = \sin(xy)$ at $\left(1, -\frac{\pi}{2}\right)$ upto 2nd degree.
3. $f(x) = e^x \cos y$ at $\left(1, \frac{\pi}{4}\right)$ upto 2nd degree.
4. $f(x) = \tan^{-1} \left(\frac{y}{x}\right)$ at $(1, 1)$ upto 2nd degree.

5. $f(x) = 4x^2 + 6y^2 + xy - 20y + x + 21$ at $(-1, 1)$.

6. $f(x) = \tan^{-1}(xy)$ at $(0.9, +1.1)$.

7. $f(x) = \sqrt{x+y}$ at $(1, 2)$.

8. $f(x) = \sin x \sin y$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$

9. $f(x) = e^x \sin y$ at $\left(-1, \frac{\pi}{4}\right)$.

Type II

[Point value is not given (i.e.) $(0, 0)$ or Maclaurin's series]

1. $f(x) = e^x \log(1+y)$ upto 3rd degree (Dec. 2017, May 2015).

2. $f(x) = \tan^{-1}\left(\frac{y}{x}\right)$ upto 3rd degree.

3. $f(x) = e^x \sin y$ upto 3rd degree.

4. $f(x) = e^x \cos y$ upto 3rd degree.

5. $f(x) = e^{2x+y}$.

6. $f(x) = e^{x-y}$.

Jacobin

Type I (two variables)

1. $x = r \cos \theta$; $y = r \sin \theta$, evaluate $\frac{\partial(x,y)}{\partial(r,\theta)}$ & $\frac{\partial(r,\theta)}{\partial(x,y)}$.

2. $x = u(1+v)$; $y = v(1+u)$ find $\frac{\partial(x,y)}{\partial(u,v)}$.

3. $u = \frac{2x-y}{2}$; $v = \left(\frac{y}{2}\right)$ find $\frac{\partial(u,v)}{\partial(x,y)}$.

4. $u = 2xy$; $v = x^2 - y^2$ & $x = r \cos \theta$; $y = r \sin \theta$ find $\frac{\partial(u,v)}{\partial(r,\theta)}$.

5. $u = 4x^2 + 6xy$; $v = 2y^2 + xy$ & $x = r \cos \theta$; $y = r \sin \theta$ find $\frac{\partial(u,v)}{\partial(r,\theta)}$.

6. Prove that $\frac{\partial(x,y)}{\partial(u,v)} \times \frac{\partial(u,v)}{\partial(x,y)} = 1$ for the following

(a) $x = u(1-v)$; $y = uv$

(b) $x = uv; \quad y = \frac{u}{v}$

(c) $x = e^v \sec u; \quad y = e^v \tan u$

(d) $x = u \cos v; \quad y = u \sin v$

Type III (3 variables)

1. Find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

(a) $u = xyz; \quad v = x^2 + y^2 + z^2; \quad w = x + y + z$

(b) $u = x + y + z; \quad uv = y + z; \quad z = uvw.$

2. Find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

(a) $u = 3x + 2y - z; \quad v = x - 2y + z; \quad w = x + 2y - z$

(b) $u = x + 3y^2 - z^3; \quad v = 4x^2yz; \quad w = 2z^2 - xy$ at $(1, -1, 0)$

(c) $u = x^2 + y^2 + z^2; \quad v = xy + yz + zx; \quad w = x + y + z$

3. Find $\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)}$

(a) $y_1 = \frac{x_2 x_3}{x_1}; \quad y_2 = \frac{x_3 x_1}{x_2}; \quad y_3 = \frac{x_1 x_2}{x_3}$

(b) $y_1 = 1 - x_1; \quad y_2 = x_1(1 - x_2); \quad y_3 = x_1 x_2(1 - x_3).$

4. If $x = r \sin \theta \cos \phi; \quad y = r \sin \theta \sin \phi; \quad z = r \cos \theta \cos \phi$ find $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$

5. if $x = \rho \cos \phi; \quad y = \rho \sin \phi; \quad z = z$ find $\frac{\partial(x,y,z)}{\partial(\rho,\phi,z)}$

6. if $x = a \cosh \phi \cos \theta; \quad y = a \sinh \phi \sin \theta$ show that $\frac{\partial(x,y)}{\partial(\phi,\theta)} = \frac{a^2}{2} [\cosh 2\phi - \cos 2\theta]$

Type III [Properties of Jacobin]

1. If $x = uv; \quad y = \frac{u+v}{u-v}$ find $\frac{\partial(u,v)}{\partial(x,y)}$.

2. If $u = x^2 - y^2; \quad v = 2xy \quad \& \quad x = r \cos \theta; \quad y = r \sin \theta$ find $\frac{\partial(u,v)}{\partial(r,\theta)}$.

3. If $u = xyz; \quad v = x^2 + y^2 + z^2; \quad w = x + y + z$ find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.

4. Verify whether the following functions are functional dependent and if so, find the relation between them

(a) $u = xy + yz + zx; \quad v = x^2 + y^2 + z^2; \quad w = x + y + z.$

(b) $u = x + y + z$; $v = x^2 + y^2 + z^2$; $w = xy + yz + zx$.

(c) $u = x - y$; $v = y - z$; $w = z - x$.

(d) $u = xe^y \sin z$; $v = xe^y \cos z$; $w = x^2 e^{2y}$.

(e) $u = x + y$; $v = x^3 + y^3$; $w = x^2 y + xy^2$.

(f) $u = \frac{x+y}{1-xy}$; $v = \tan^{-1} x + \tan^{-1} y$.

Maxima and Minima (or) Extreme Values

❖ **Examine the following functions for extreme values**

1. $f(x, y) = xy + \frac{9}{x} + \frac{3}{y}$.

2. (a) $f(x, y) = x^2 + y^2 + 6y + 12$ (b) $f(x, y) = x^2 + y^2 + 6y + 12$

3. $f(x, y) = \sqrt{1 - x^2 - y^2}$.

4. $f(x, y) = 3x^2 - y^2 + x^3$.

5. $f(x, y) = x^3 + y^3 - 3xy$.

6. $f(x, y) = x^3 + y^3 - 3axy$.

7. $f(x, y) = x^3 + y^3 - 12x - 3y + 20$ (Model Dec. 2017)

8. $f(x, y) = x^3 y^2 (1 - x - y)$.

9. $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$ (Model Jan 2016).

10. $f(x, y) = x^4 + 2x^2 y - x^2 + 3y^2$.

Constrained Maxima and Minima (or) Lagrange's method of Multipliers

1. Find the min. value of $x^2 + y^2 + z^2$, given that $ax + by + cz = p$.

2. Find the min. value of $x^2 + y^2 + z^2$, given that $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$.

3. Find the volume of the largest rectangular solid which can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ (Model May 2015, Apr./May 2018).

4. Show that rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

5. Find the shortest and longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.

6. Find the maximum and minimum distance of the point $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$. (May 2016)

7. A thin closed rectangular box is to have one edge equal to twice the other, and a constant volume 72m^3 . Find the least surface area of the box.
8. A rectangular box, open at the top, is to have a volume of 31 cc. Find the dimensions of the box, which requires the least material for its construction. (Dec. 2017)
9. Find the dimensions of the rectangular box without a top of maximum capacity, whose surface is 108 sq.cm.
10. The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.